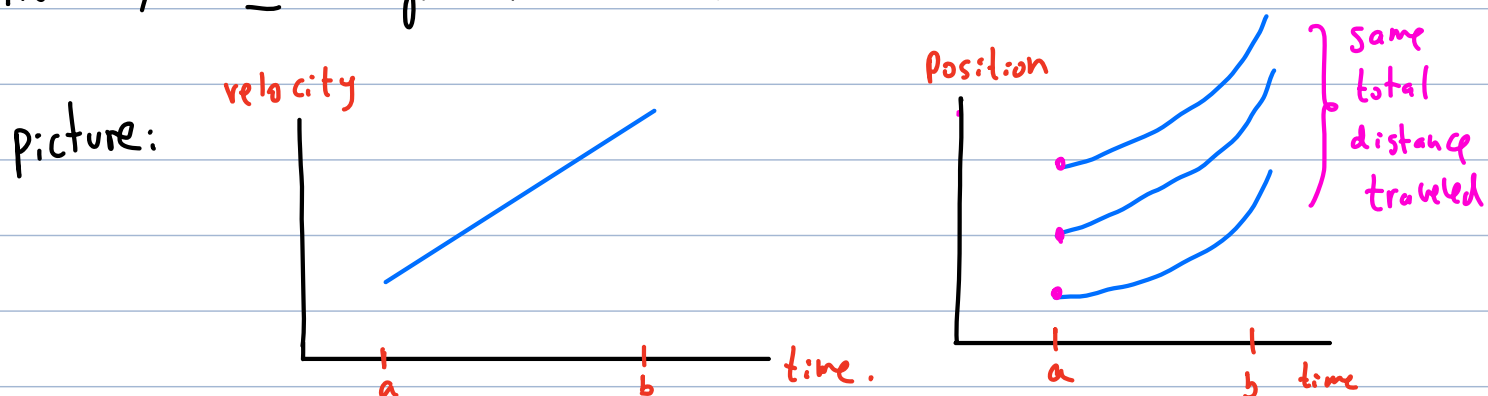


Calc I Lecture 23: Areas, distances and the definite integral

The distance problem

We've seen that if we know the velocity function, this is not enough information to deduce the position.

However, it is enough information to deduce the distance traveled.



But how do we actually go about finding the total distance traveled?

Approach 1: using antiderivatives

$v(t) \Rightarrow$ find an antiderivative of v , $F(t)$

$$s(t) = F(t) + \underbrace{C}_{\text{unknown}}$$

$$\text{"displacement"} = s(b) - s(a) = (F(b) + \cancel{C}) - (F(a) + \cancel{C}) = \underbrace{F(b) - F(a)}$$

Approach 2:

we can approximate the answer
by adding up a lot of "small distances"

Example Suppose we are given the following table of your car's speed at various points in time:

Time (s)	0	5	10	15	20	25	30
Velocity (ft/s)	25	31	35	43	47	<u>46</u>	41

How would you approximate the total distance covered over these 30 seconds?

Answer: approximate distance over each interval,
then add up the estimates

$$\text{speed} = \frac{\text{distance}}{\text{time}} \Rightarrow \text{distance} = \text{speed} \cdot \text{time}$$

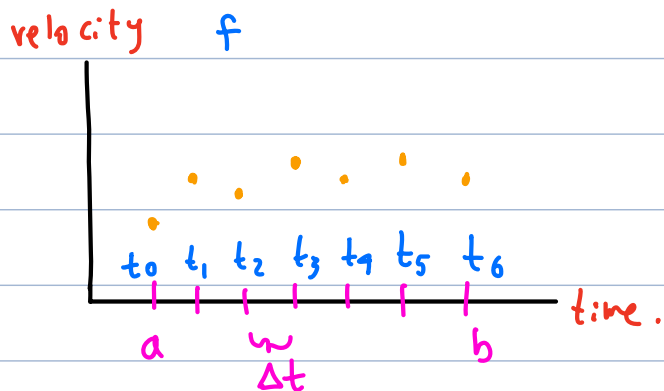
between $t=0$ and $t=5$: distance $\approx 25 \cdot 5$
(use "left endpoints")

$$\begin{aligned} \text{Overall: } & 25 \cdot 5 + 31 \cdot 5 + 35 \cdot 5 + 43 \cdot 5 + 47 \cdot 5 + 46 \cdot 5 \\ \text{(using left endpoints)} & = \boxed{1135 \text{ ft}} \text{ approximate distance} \end{aligned}$$

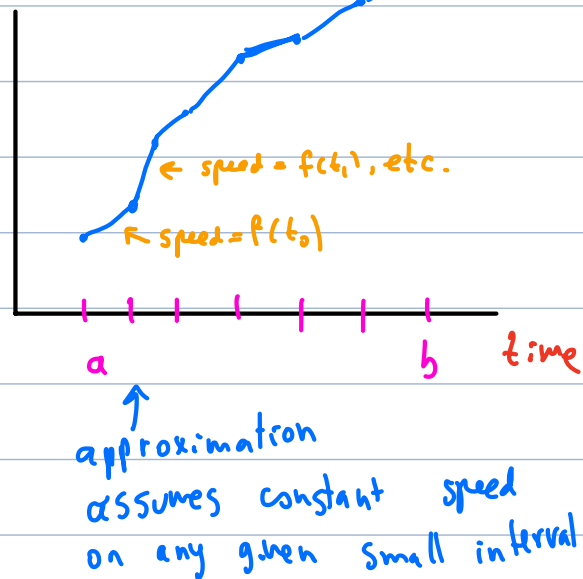
using right endpoints:

$$31 \cdot 5 + 35 \cdot 5 + 43 \cdot 5 + 47 \cdot 5 + 46 \cdot 5 + 41 \cdot 5$$

Picture:



Position



More general situation:

terminology:

- n = number of subintervals that we divide $[a, b]$ into. (in our example $n=6$)
- $\Delta t = \frac{b-a}{n}$ (length of each subinterval)
- we label the endpoints of the intervals
 $a = t_0, t_1, t_2, \dots, t_n = b$
- let $f(t)$ represent velocity as a function of time
- For each i ^{i ranges from 1 to n} i _{th} interval, approximate velocity by
 - if left endpoints: $f(t_{i-1})$
 - if right endpoints: $f(t_i)$

• distance covered over i th interval:

if left: $f(t_{i-1}) \cdot \Delta t$

if right: $f(t_i) \cdot \Delta t$

• Formula for n th approximation:

if left: $f(t_0) \cdot \Delta t + f(t_1) \cdot \Delta t + \dots + f(t_{n-1}) \cdot \Delta t$

if right: $f(t_1) \cdot \Delta t + f(t_2) \cdot \Delta t + \dots + f(t_n) \cdot \Delta t$

Formula for the true distance:

$$d = \lim_{n \rightarrow \infty} \left(f(t_1) \cdot \Delta t + f(t_2) \cdot \Delta t + \dots + f(t_n) \cdot \Delta t \right)$$

Summation notation:

$f(t_1) \cdot \Delta t + f(t_2) \cdot \Delta t + \dots + f(t_n) \cdot \Delta t$ } i th term for general i is $\underbrace{f(t_i) \cdot \Delta t}$

$$= \sum_{i=1}^n f(t_i) \cdot \Delta t$$

Greek letter Σ (sigma) stands for "sum"

in words:
add up all terms of the form $f(t_i) \cdot \Delta t$ where i ranges from $i=1$ to $i=n$

Using summation notation:

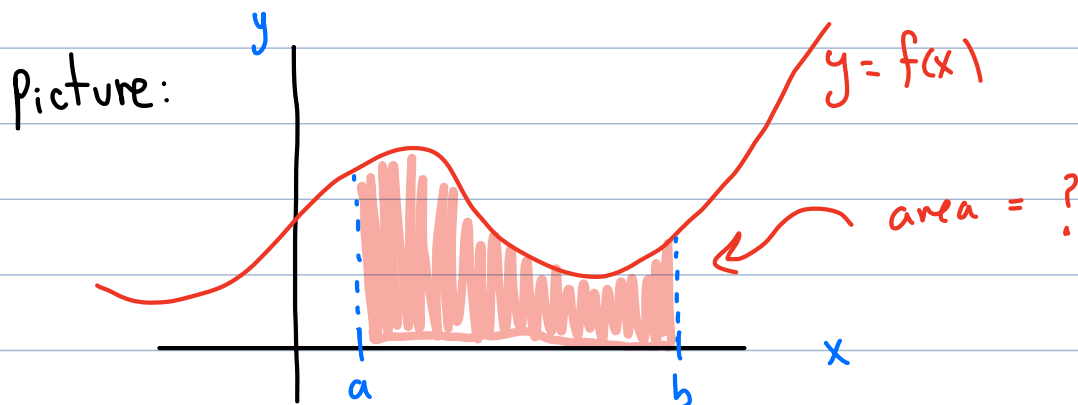
$$\text{distance} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_i) \cdot \Delta t$$

"divide time interval into n pieces, estimate distance over each piece, add up all the small distances, take limit as $n \rightarrow \infty$ "

The area problem

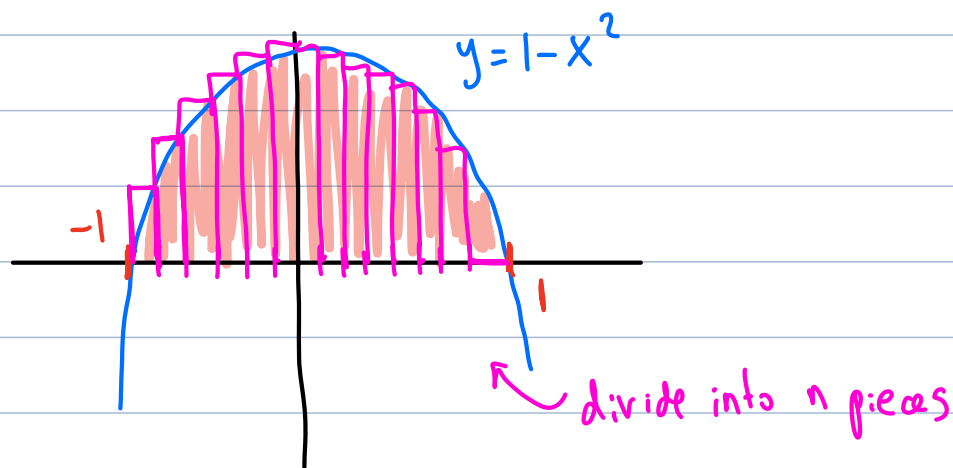
Next we turn to the area problem.

Suppose we are given a continuous function $f(x)$ (assume non-negative) and consider the region S bounded by the graph of $y=f(x)$, the x -axis and by $x=a$ and $x=b$.



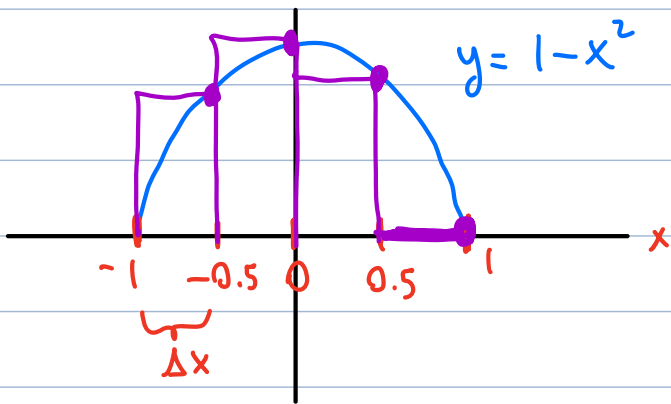
Question: what is the area of S ?

Example: estimate the area under the parabola $y=1-x^2$ between $x=-1$ and $x=1$



Idea: as with distance, partition into a bunch of intervals, estimate area over each interval, add up estimates.

e.g. with $n=4$, and right endpoints:



$$\Delta X = \frac{b-a}{n} = \frac{1-(-1)}{4} = \frac{2}{4} = 0.5$$

for each rectangle: $\text{area} = \underbrace{\Delta x}_{\substack{|| \\ 0.5}} \cdot \underbrace{\text{height}}_{1 - x_i^2}$

total area: $0.5 \cdot (1 - (-0.5)^2) + 0.5 \cdot (1 - 0^2) + 0.5(1 - 0.5^2) + 0$

$= \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{3}{4} + 0 = \frac{3}{8} + \frac{1}{2} + \frac{3}{8} = \frac{10}{8} = 1.25$

General picture:

area under the curve $y = f(x)$ between a and b

- divide interval $[a, b]$ into n parts

- each part will have length $\Delta x = \frac{b-a}{n}$

- use left or right endpoints

- if right end point:

area of i^{th} rectangle will be $\underbrace{\text{width}}_{\Delta x} \cdot \underbrace{\text{height}}_{f(x_i)}$

$\hookrightarrow$ right endpoint

- total estimate: $\Delta x \cdot f(x_1) + \Delta x \cdot f(x_2) + \Delta x \cdot f(x_3) + \dots + \Delta x \cdot f(x_n)$

- true area under curve: $A = \lim_{n \rightarrow \infty} (\Delta x \cdot f(x_1) + \Delta x \cdot f(x_2) + \dots + \Delta x \cdot f(x_n))$

Observation: the formula for distance and area looks the same!

How come?

Both have to do with "adding up lots of little bits" of a function

We can unite both of these problems under the framework of integrals.

Definite integrals

Let $f(x)$ be a continuous function. Let $[a, b]$ be an interval.

Definition: An n_{th} Riemann sum of f over the interval $[a, b]$

is a sum $\Delta x \cdot f(x_1^*) + \Delta x \cdot f(x_2^*) + \dots + \Delta x \cdot f(x_n^*)$

same thing
in
summation
notation

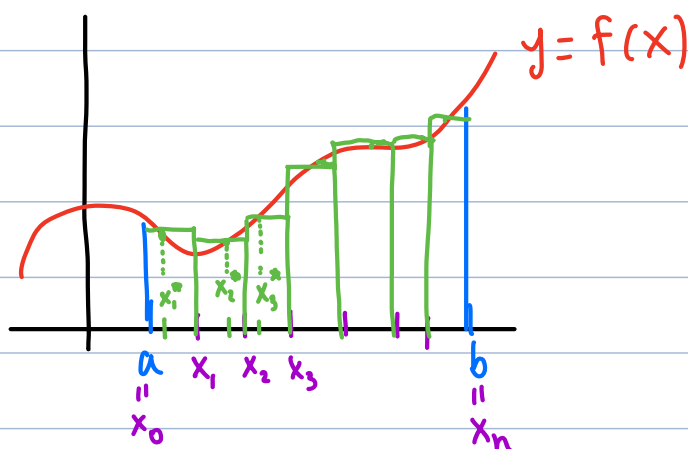
$$\Rightarrow \sum_{i=1}^n \underbrace{\Delta x}_{\text{length}} \cdot \underbrace{f(x_i^*)}_{\text{height}}$$

where $\Delta x = \frac{b-a}{n}$

and for each i , x_i^* is a representative point of i th interval
(left endpoint, right endpoint
or anything in between)

$$\underbrace{x_{i-1}}_{\text{left endpoint}} \leq x_i^* \leq \underbrace{x_i}_{\text{right endpoint}}$$

Picture:



Definition: A function f is integrable over $[a, b]$

if all Riemann sums approach the same limiting value as $n \rightarrow \infty$ regardless of our choice of representative points x_i^* .

In this case, we write

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f(x_i^*) \cdot \Delta x \right)$$

in words:

"the integral of $f(x)$
from a to b "

↑
limit of the Riemann
sums as $n \rightarrow \infty$

Some special Riemann sums:

- left endpoint: choose $x_i^* = x_{i-1}$
- right endpoint: choose $x_i^* = x_i$
- midpoint: choose $x_i^* = \bar{x}_i$ (point halfway between right and left)

Example: use each of these options to estimate

$$\int_0^1 e^x dx \quad \text{using } \underline{n=5}$$

For each one, is it an over-approximation or under-approximation?



$$\Delta x = \frac{b-a}{n} = \frac{1-0}{5} = \frac{1}{5} = 0.2$$

$$f(x) = e^x$$

• left-endpoint:

$$\begin{aligned} \int_0^1 e^x dx &\approx \underline{\Delta x} f(0) + \underline{\Delta x} \cdot f(0.2) + \underline{\Delta x} \cdot f(0.4) + \underline{\Delta x} \cdot f(0.6) + \underline{\Delta x} \cdot f(0.8) \\ &= \Delta x (f(0) + f(0.2) + f(0.4) + f(0.6) + f(0.8)) \\ &= 0.2 (e^0 + e^{0.2} + e^{0.4} + e^{0.6} + e^{0.8}) \approx \boxed{1.552} \end{aligned}$$

• right-endpoint:

$$\int_0^1 e^x dx \approx 0.2 (e^{0.2} + e^{0.4} + e^{0.6} + e^{0.8} + e^1) \approx \boxed{1.896}$$

• midpoint:

$$\int_0^1 e^x dx \approx \underline{0.2} \left(\underline{e^{0.1}} + \underline{e^{0.3}} + \underline{e^{0.5}} + \underline{e^{0.7}} + \underline{e^{0.9}} \right) \approx \boxed{1.715}$$

$\Delta x \quad f(\bar{x}_1) \quad f(\bar{x}_2) \quad \dots$

over/under

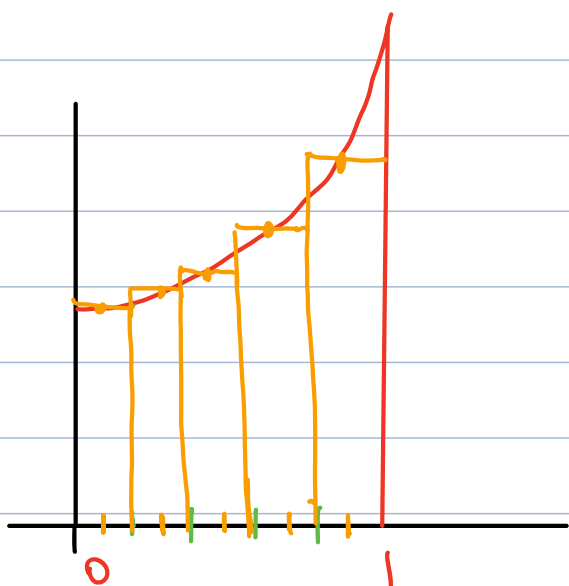


left endpoint:

rectangles below graph
under approximation

right endpoint:

rectangles above graph:
over approximation



midpoint:

because e^x is concave up,

area below curve $>$ area above curve

$\Rightarrow$ underestimate

General observation

- if a function is increasing,

left endpoint approx is an underestimate

right endpoint approx is an overestimate

- if a function is decreasing

left endpoint approx is an overestimate

right endpoint approx is an underestimate



- if a function is concave up

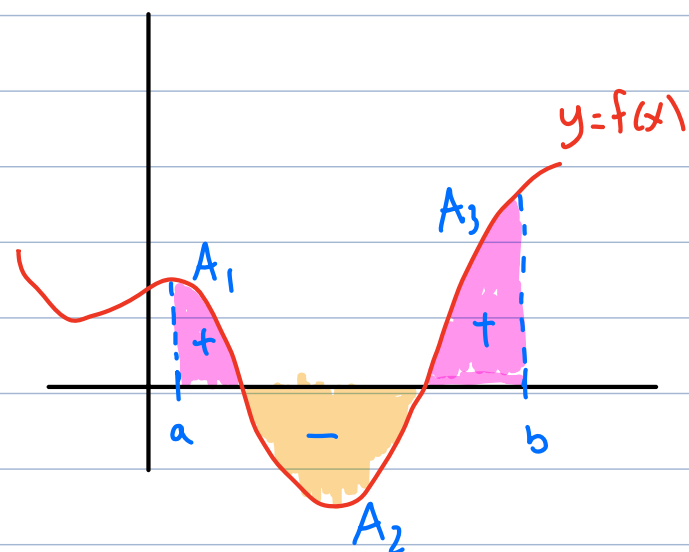
midpoint approx is an underestimate

- if a function is concave down

midpoint approx is an overestimate

What if f is negative at some points?

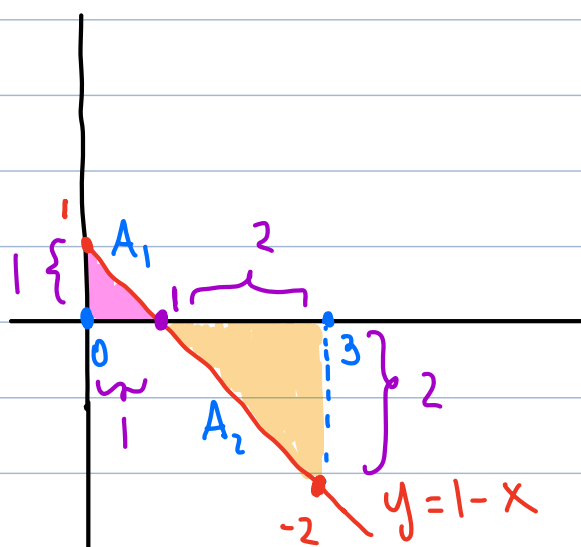
Then we can interpret $\int_a^b f(x) dx$ as "net area" between curve and x-axis



$$\int_a^b f(x) dx = A_1 + A_3 - A_2$$

"net area" = (area below curve above x axis)
 - (area below x-axis above curve)

Example: use geometry to find $\int_0^3 (1-x) dx$



$$\begin{aligned} &= A_1 - A_2 \\ &= \frac{1 \cdot 1}{2} - \frac{2 \cdot 2}{2} = \frac{1}{2} - 2 = \boxed{-\frac{3}{2}} \end{aligned}$$

x-intercept: $0=1-x \Rightarrow x=1$

- in the context of velocity, if $f(t)$ represents velocity
 $\int_a^b f(t) dt$ represents

displacement = net change in position

$$= (\text{distance going right}) - (\text{distance going left})$$

Recognizing sums as Riemann sums

Formula for definite integral (using right endpoints)

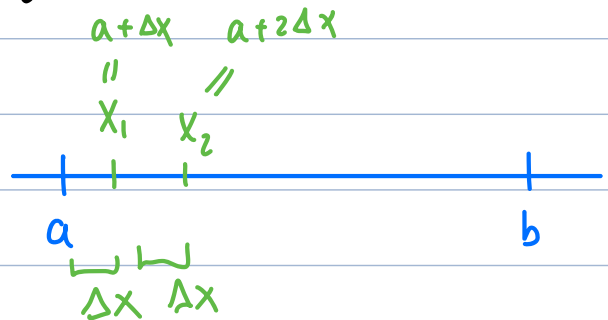
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \underbrace{f(x_i) \Delta x}_{\text{Riemann sum with right endpoints}} \right)$$

More concrete formula for integral:

$$\bullet \Delta x = \frac{b-a}{n}$$

$$\bullet x_i = a + i \Delta x$$

$$x_i = a + i \left(\frac{b-a}{n} \right)$$



so the above formula becomes:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \underbrace{f\left(a + i \left(\frac{b-a}{n} \right)\right)}_{x_i} \underbrace{\left(\frac{b-a}{n} \right)}_{\Delta x} \right)$$

Example: Write as a limit expression:

$$\int_{-1}^2 \sin(2x+1) dx$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f(x_i) \Delta x \right)$$

More specifically,

$$a = -1, b = 2, f(x) = \sin(2x+1)$$

$$\Delta x = \frac{b-a}{n} = \frac{2-(-1)}{n} = \frac{3}{n}$$

$$x_i = a + i\Delta x = -1 + \frac{i \cdot 3}{n} = \underbrace{-1 + \frac{3i}{n}}_{x_i}$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \sin(2 \underbrace{x_i}_{=}) + 1) \cdot \frac{3}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \underbrace{\sin\left(2 \left(-1 + \frac{3i}{n}\right) + 1\right)}_{f(x_i)} \cdot \underbrace{\frac{3}{n}}_{\Delta x} \right)$$

Example Express as a definite integral:

a) (easy mode) $\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^3 + x_i \sin x_i) \cdot \Delta x$ over $[0, \pi]$

b) (hard mode) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{1}{1 + (i/n)^2}$

a) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \underbrace{(x_i^3 + x_i \sin x_i)}_{f(x_i)} \cdot \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n \underbrace{f(x_i)}_{f(x)} \Delta x = \int_a^b f(x) dx$

$\Rightarrow f(x) = x^3 + x \sin(x)$

$a = 0, b = \pi$

$= \int_0^{\pi} (x^3 + x \sin(x)) dx$

b) $\lim_{n \rightarrow \infty} \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \frac{1}{1 + (i/n)^2} \right)}_{f(x_i) \Delta x} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \int_a^b f(x) dx$

try: $\Delta x = \frac{1}{n}$

$\frac{1}{1 + (i/n)^2} = f(x_i) ?$

$\underbrace{x_i = \frac{i}{n}}_{f(x)} \Rightarrow \underbrace{f(x) = \frac{1}{1 + x^2}}$

$$x_i = a + i \cdot \Delta x = a + i \cdot \frac{(b-a)}{n}$$

$$\stackrel{?}{=} \frac{i}{n} \Rightarrow a=0, \underline{b-a=1} \Rightarrow a=0, b=1$$

guess: $= \int_0^1 \frac{1}{1+x^2} dx$

double check:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$$

$$x_i = a + i \Delta x = 0 + i \cdot \frac{1}{n} = \underbrace{\frac{i}{n}}_{x_i}$$

$$f(x_i) = \frac{1}{1+x_i^2} = \frac{1}{1+(\frac{i}{n})^2}$$

$$\sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \frac{1}{1+(\frac{i}{n})^2} \cdot \frac{1}{n}$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{1}{1+(\frac{i}{n})^2} \quad \checkmark$$

Example: write as an integral:

$$\lim_{n \rightarrow \infty} \underbrace{\frac{3}{n}}_{\Delta x} \cdot \sum_{i=1}^n \underbrace{\ln(2 + 3 \cdot \frac{i}{n})}_{f(x_i)} = \int_a^b f(x) dx$$

↑

find $a, b, f(x)$

$$x_i = a + i \cdot \Delta x$$

$$= \underline{a} + i \cdot \frac{\overbrace{(b-a)}^?}{n} = \underline{2} + \overbrace{3}^? \cdot \frac{i}{n}$$

pattern match: $a=2$

$$b-a=3 \Rightarrow \underline{b=3+a=5}$$

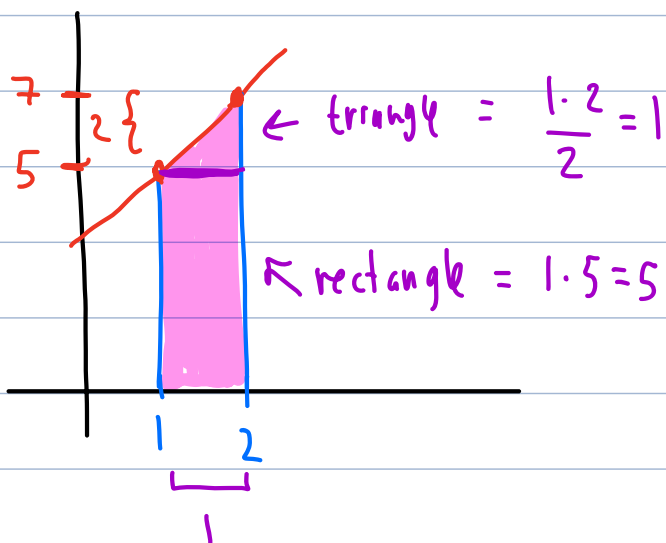
$$x_i = 2 + 3 \cdot \frac{i}{n}$$

$$\ln(\overbrace{2 + 3 \cdot \frac{i}{n}}^{x_i}) = f(x_i) \Rightarrow \underline{f(x) = \ln(x)}$$

$$= \int_2^5 \ln(x) dx$$

Evaluating certain simple integrals by recognizing them as areas:

e.g, what is $\int_1^2 (2x+3) dx = 6$

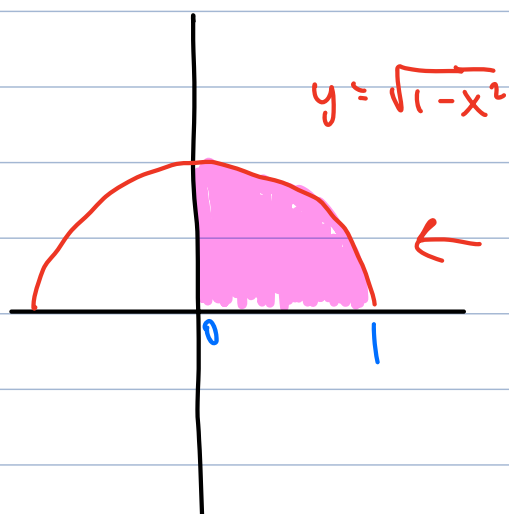


total area = $5 + 1 = 6$

(alternatively,
area of trapezoid =
 $1 \cdot \left(\frac{5+7}{2}\right) = 6$)

e.g, $\int_0^1 \sqrt{1-x^2} dx =$

area under the curve $y = \sqrt{1-x^2} \Rightarrow y^2 = 1-x^2 \Leftrightarrow x^2 + y^2 = 1$
y positive circle radius 1



quarter circle, area = $\frac{1}{4} \cdot (\pi \cdot 1^2)$
 $= \boxed{\frac{\pi}{4}}$